From Noisy MPTS Points to Positive Plasma Profiles
A Pedagogical Companion to the Bayesian Profile Lab

UW Plasma Hackathon Project

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Technical abstract. The bundled NSTX multipoint Thomson-scattering file already contains processed electron-temperature and electron-density measurements, their reported standard uncertainties, and the major radius of each spatial channel. The project therefore solves a noisy-function reconstruction problem. For either physical quantity $f \in \{T_e, n_e\}$, a dimensionless latent field $z : I \to \mathbb{R}$ receives a stationary Matérn-5/2 Gaussian-process prior, the physical profile is constructed as $f(R) = f_{\text{ref}} \exp z(R) > 0$, and the reported uncertainty remains an additive Gaussian likelihood in keV or cm^{-3}. A Laplace approximation at each of nine correlation-length quadrature nodes supplies an interactive approximation to the non-Gaussian posterior. Analytic kernel derivatives produce correlated samples of z and z', from which the program computes f', $d \log f/dR$, and the reciprocal scale length. A sign-identification gate withholds the reciprocal wherever the posterior supports a zero gradient. The implementation completed all 1,486 temperature and density fits in the bundled sweep, eliminated the 590 negative-median profiles produced by the original linear-space baseline, and passed compact synthetic calibration tests. Its intervals remain conditional on the stated stationary-kernel and likelihood family, its radial coordinate is geometric major radius, and its Laplace/quadrature posterior still requires comparison against full posterior sampling before scientific deployment.

1 The measurement already exists

The governing distinction appears before any regression equation. NSTX-U's diagnostic analysis converts calibrated optical signals into local estimates of T_e and n_e. The hackathon file begins after that conversion. One selected shot and acquisition time supplies

$$\mathcal{D}_T = \{(R_i, T_{e,i}, \sigma_{T,i})\}_{i=1}^{N_T}, \tag{1}$$

$$\mathcal{D}_n = \{(R_i, n_{e,i}, \sigma_{n,i})\}_{i=1}^{N_n}. \tag{2}$$

Here R_i is the geometric machine major radius of channel i; $\sigma_{T,i}$ and $\sigma_{n,i}$ are the standard uncertainties reported by the processed diagnostic. The inference target is a pair of positive functions on the measured interval $I = [R_{\min}, R_{\max}]$:

$$(T_e, n_e) \in C^1(I, \mathbb{R}_{>0}) \times C^1(I, \mathbb{R}_{>0}). \tag{3}$$

The first derivative is included in the target space because the requested outputs contain df/dR and a normalized gradient.

The HDF5 reader found 14 shots and 743 acquisition times. Each acquisition contains 30 radii and therefore defines two independent profile problems, giving $2 \times 743 =$ 1,486 fits in the complete sweep. In the default case, shot 141687 at 0.3150 s, the channels span 27.60–156.17 cm with a median separation of 2.916 cm. The median reported relative uncertainties are 4.94% for temperature and 4.47% for density.

Figure 1 fixes the boundary between plasma-diagnostic physics and the reconstruction performed here. The upstream NSTX-U pipeline evaluates a relativistic Selden spectrum through the measured polychromator response, adjusts temperature and amplitude by a spectral chi-squared fit, and converts the calibrated amplitude into density [1]. A compact channel model is

$$\mu_j = b_j + A E_L n_e \int \mathcal{T}_j(\lambda) \eta_j(\lambda) S_\lambda^{\text{Selden}}(\lambda; T_e, \theta)\, \mathrm{d}\lambda. \tag{4}$$

The expected signal μ_j in optical channel j combines background b_j, laser energy E_L, density n_e, filter transmission $\mathcal{T}_j$, detector efficiency η_j, and the temperature-dependent spectral shape. Equation (4) supplies physical provenance. No variable on its left-hand side is present in the bundled HDF5 file, so the app does not evaluate this equation.

2 A positive function from uncertain points

Choose one quantity $q \in \{T, n\}$ and write its processed observations as y_i with standard uncertainties σ_i. The measurement model is heteroscedastic: each channel carries its own noise variance,

$$y_i = f(R_i) + \epsilon_i, qquad \epsilon_i \sim \mathcal{N}(0, \sigma_i^2). \tag{5}$$

Consequently, a small-σ_i channel contributes greater likelihood curvature than a large-σ_i channel. The optional relative-uncertainty filter changes which observations enter the posterior; it is a sensitivity control, since Equation (5) already down-weights a finite high-uncertainty point.

Temperature and density occupy $\mathbb{R}_{>0}$. We encode that support through a latent logarithmic coordinate. Let $f_{\text{ref}} >$ 0 be a fixed physical scale: 1 keV for T_e and 10^{13} cm^{-3} for n_e. Construct

$$\begin{aligned} z(R) &= \beta + u(R), & u &\sim \mathcal{GP}(0, k_{5/2}), \\ f(R) &= f_{\text{ref}} \exp z(R), & y_i \mid z &\sim \mathcal{N}\left(f_{\text{ref}} e^{z(R_i)}, \sigma_i^2\right). \end{aligned} \tag{6}$$

The latent field $z : I \to \mathbb{R}$ is dimensionless; exponentiation maps it into a positive physical profile. The likelihood in Equation (6) remains Gaussian in the original measurement units. Thus positivity of the latent physical state does not replace the diagnostic's additive error statement with a lognormal observation model.

The constant β sets the log-profile level. The present priors are

$$\begin{aligned} \beta &\sim \mathcal{N}(0, 2^2), & \sigma_z &\sim \text{HalfNormal}(1), \\ \log(\ell/D) &\sim \mathcal{N}(\log 0.12, 0.75^2). \end{aligned} \tag{7}$$

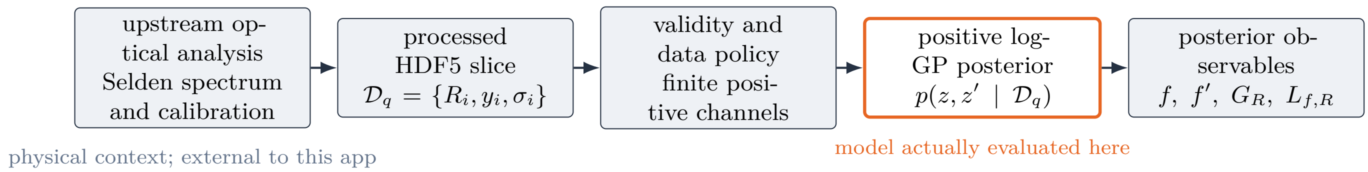

Figure 1: The calculation map. The upstream Thomson-spectrum fit explains the origin of the measurements; the repository begins with processed profile points and propagates their uncertainty through a positive spatial model.

where $D = R_{\max} - R_{\min}$, σ_z is the latent covariance amplitude, and ℓ is its radial correlation length. The code rescales radius to $x = (R - R_{\min})/D \in [0, 1]$ for computation, then restores physical derivative units through $d/dR = D^{-1}d/dx$.

2.1 Smoothness as a scientific prior

For $d = |R - R'|$ and $a = \sqrt{5}/\ell$, the default covariance is

$$k_{5/2}(R, R') = \sigma_z^2 \left(1 + ad + \frac{a^2d^2}{3}\right) e^{-ad}. \tag{8}$$

The correlation length ℓ defines the radial scale over which latent profile values inform one another. A short ℓ admits local structure; a long ℓ couples distant channels into a broad profile. The Matérn-5/2 family supplies a regular first-derivative process while retaining less prior smoothness than the radial-basis-function kernel. The implementation also prevents inferred correlation lengths below the typical channel separation, because no observation pattern can resolve repeatable sub-channel structure.

The project fits T_e and n_e independently. This avoids asserting an unvalidated cross-quantity covariance. It also fits each acquisition time independently, so rapid temporal changes are neither pooled nor smoothed across frames.

3 The posterior moves through derivatives

Because $f = f_{\text{ref}}e^z$, Equation (6) is nonlinear in the Gaussian field z. Let $\mathbf{z} = (z(R_1), \ldots, z(R_N))^\top$ and $K_{XX} = [k(R_i, R_j)]_{ij}$. Apart from constants, the negative log posterior at fixed hyperparameters is

$$\begin{aligned}\Phi(\mathbf{z}, \beta, \sigma_z; \ell) = {}& \frac{1}{2}\sum_{i=1}^{N} \frac{(y_i - f_{\text{ref}}e^{z_i})^2}{\sigma_i^2} + \frac{1}{2}\sum_{i=1}^{N} \log \sigma_i^2 \\ &+ \frac{1}{2}(\mathbf{z} - \beta\mathbf{1})^\top K_{XX}^{-1}(\mathbf{z} - \beta\mathbf{1}) \\ &+ \frac{1}{2}\log|K_{XX}| + \Phi_{\text{prior}}.\end{aligned} \tag{9}$$

Here Φ_{prior} contains Equation (7). The optimized variables $\mathbf{z}$, β, and σ_z remain inside the posterior approximation; they have not been replaced by a smoothed curve.

Interactive inference proceeds through a finite mixture:

1. Place nine quadrature nodes in $\log(\ell/D)$ between the channel-spacing floor and the prior-supported upper range.
2. At each node m, minimize Φ with L-BFGS-B and compute the local Hessian $H_m = \nabla^2\Phi(\widehat{\vartheta}_m)$.
3. Form a stabilized Laplace covariance H_m^{-1} and an evidence weight

$$\log w_m \doteq -\Phi(\widehat{\vartheta}_m) - \frac{1}{2}\log|H_m| + \log p(\log \ell_m) + \log \Delta_m. \tag{10}$$

4. Draw latent and hyperparameter points from the normalized mixture, then draw conditional functions and derivatives on a dense radial grid.

The symbol $\doteq$ marks a Laplace/quadrature approximation. Integration over several length scales carries materially more posterior volume information than one optimized empirical-Bayes length, yet it remains an approximation to full Bayesian sampling.

3.1 Analytic differentiation preserves correlation

For $\delta = R - R'$ and $d = |\delta|$, differentiating Equation (8) gives

$$\text{Cov}[z(R), z'(R')] = \frac{\sigma_z^2a^2}{3}\,\delta(1 + ad)e^{-ad}, \tag{11}$$

$$\text{Cov}[z'(R), z'(R')] = \frac{\sigma_z^2a^2}{3}\left(1 + ad - a^2d^2\right)e^{-ad}. \tag{12}$$

These are covariances, not finite-difference recipes. They construct one joint random object containing profile values and radial derivatives [4]. At evaluation radii $\mathbf{R}_*$, stack

$$\mathbf{d}_* = \begin{pmatrix}\mathbf{z}_* \\ \mathbf{g}_*\end{pmatrix}, \qquad \mathbf{g}_* = (z'(R_{*,1}), \ldots, z'(R_{*,M}))^\top. \tag{13}$$

Conditional on one posterior draw of the training latent values and kernel parameters,

$$\mathbf{d}_* \mid \mathbf{z}_X, \phi \sim \mathcal{N}(\boldsymbol{\mu}_d, C_d), \tag{14}$$

$$\boldsymbol{\mu}_d = \mathbf{m}_d + K_{dX}K_{XX}^{-1}(\mathbf{z}_X - \mathbf{m}_X), \tag{15}$$

$$C_d = K_{dd} - K_{dX}K_{XX}^{-1}K_{Xd}. \tag{16}$$

The off-diagonal blocks retain the dependence between z and z'. Every derived posterior sample therefore represents one internally consistent profile and gradient.

4 One sample, four physical observables

Let $(z^{(s)}, z'^{(s)})$ be sample s from Equation (14). The samplewise map is

$$f^{(s)}(R) = f_{\mathrm{ref}} e^{z^{(s)}(R)}, \tag{17}$$

$$f'^{(s)}(R) = f^{(s)}(R) z'^{(s)}(R), \tag{18}$$

$$G_R^{(s)}(R) = \frac{f'^{(s)}(R)}{f^{(s)}(R)} = z'^{(s)}(R), \tag{19}$$

$$L_{f,R}^{(s)}(R) = \frac{f^{(s)}(R)}{f'^{(s)}(R)} = \frac{1}{z'^{(s)}(R)}. \tag{20}$$

The profile has the physical units of f; f' carries those units per cm; $G_R = d \log f / dR$ has units cm^{-1}; and $L_{f,R}$ has units cm. Pointwise medians and selected quantiles across $s = 1, \ldots, S$ form the displayed central curves and credible intervals. They are pointwise posterior bands: at each radius they contain the selected posterior mass under this model. They are not simultaneous whole-function bands or repeated-experiment confidence guarantees.

The reciprocal in Equation (20) creates a real identifiability boundary. At confidence level c, define

$$p_+(R) = \Pr[z'(R) > 0 \mid \mathcal{D}], \qquad \alpha = \frac{1-c}{2}. \tag{21}$$

The gradient sign is unresolved when $\alpha < p_+(R) < 1 - \alpha$. In that region the posterior crosses the pole at $z' = 0$, so a finite scale length would hide the defining uncertainty. The app reports G_R and withholds $L_{f,R}$. It also withholds the scale length within two median channel spacings of either measurement boundary, where a derivative is constrained from only one side.

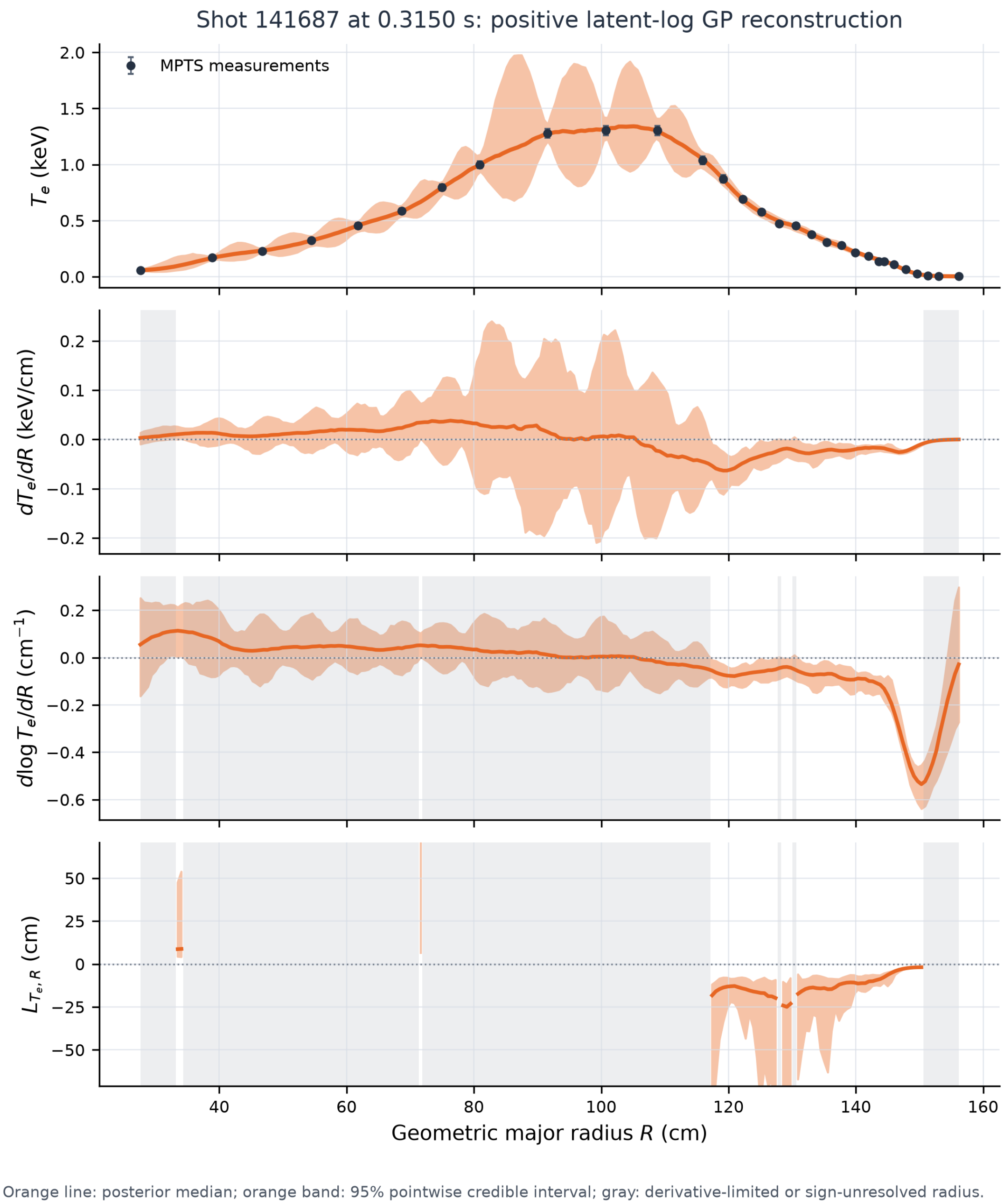

Figure 2: The implemented posterior map for the default temperature case. The same correlated samples generate all four panels. The gray regions mark one-sided derivative boundaries or radii where the 95% posterior interval does not identify the sign of $d\log T_e/dR$; the reciprocal scale length is therefore omitted. The broad profile can remain well measured even where its local derivative is weakly identified.

Figure 2 exposes a central lesson of the project. Profile precision does not transfer automatically to derivative precision. Near a broad maximum, many positive functions pass through almost the same measured values while possessing different small slopes. Their ratio f/f' is the least stable exactly where the profile appears visually flat. Refusing to draw a finite reciprocal in that region is part of the inference, not a plotting failure.

5 Major radius is not an outward flux coordinate

The file supplies geometric major radius R. Along the NSTX-U horizontal diagnostic chord, increasing R moves toward the magnetic axis on the inboard branch and away from it on the outboard branch [1]. Hence the sign of $d\log f/dR$ does not encode one globally outward direction.

If an equilibrium reconstruction supplies a flux coordinate $\rho(R)$, then each monotone branch obeys

$$\frac{d\log f}{d\rho} = \frac{d\log f/dR}{d\rho/dR}. \tag{22}$$

The denominator changes sign across the magnetic axis. A transport quantity such as $a/L_f = -a\,d\log f/dr$ therefore requires an equilibrium map, an outward minor-radius or flux convention, a normalization length a, and propagation of mapping uncertainty. The current GUI labels its result $d\log f/dR$ in cm^{-1} so the coordinate claim remains exact.

6 Scientific controls, not cosmetic sliders

Each interface control changes a declared element of the inference or its reporting policy:

Control	Mathematical role
Shot, time, quantity	Selects one data set $\mathcal{D}_q$; T_e and n_e remain independent.
Length scale	Sets or marginalizes the kernel scale ℓ.
Credible level	Sets c and the sign gate $\alpha = (1-c)/2$.
Samples	Controls Monte Carlo resolution of transformed quantiles.
Edge point	Adds a distinct likelihood term beyond the final measured radius.
Relative-error filter	Removes channels for a stated sensitivity analysis.
Bottom panel	Displays G_R, p_+, or the gated reciprocal $L_{f,R}$.

No edge condition is active by default. If independent physics supplies $R_{\mathrm{edge}} > R_{\max}$ and $f_{\mathrm{edge}} \pm \sigma_{\mathrm{edge}}$, the program adds

$$y_{\mathrm{edge}} \mid z \sim \mathcal{N}\left(f_{\mathrm{ref}} e^{z(R_{\mathrm{edge}})}, \sigma_{\mathrm{edge}}^2\right). \tag{23}$$

Requiring a distinct coordinate prevents a duplicate observation at the last channel from being mislabeled as a boundary condition.

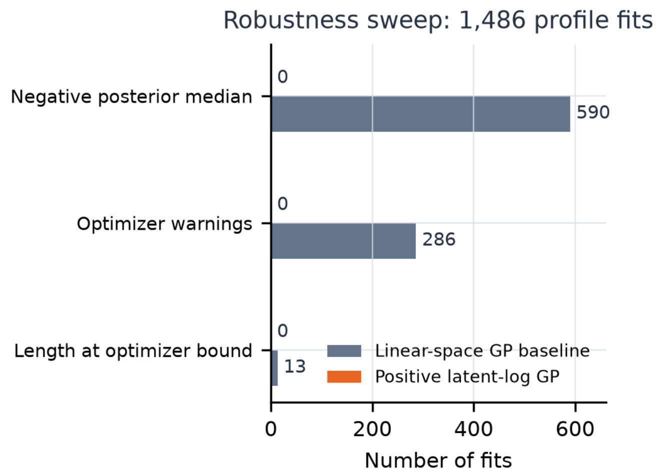

Figure 3: Numerical failure indicators from the bundled-profile sweeps. The positive latent-log model removes negative physical support by construction and did not trigger the optimizer-warning or boundary counters used in this compact sweep. The two sweeps used their respective hackathon inference implementations; the bars diagnose failure modes, not equal-cost runtime.

The provisional validity gate first collects finite positive pairs (y_i, σ_i). It then computes

$$m_{\mathrm{rel}} = \mathrm{median}_i\, \frac{\sigma_i}{y_i}, \qquad N_2 = \#\{i : y_i/\sigma_i > 2\}. \tag{24}$$

The profile is labeled unusable when $m_{\mathrm{rel}} > 1$ or $N_2 < 6$, cautionary when $m_{\mathrm{rel}} > 0.5$ or $N_2 < 10$, and usable otherwise. These thresholds separate obvious absent-signal acquisitions in the hackathon file. The HDF5 schema lacks upstream saturation and analysis-status flags, so the labels are a transparent provisional policy rather than a calibrated diagnostic-quality classifier.

7 The model changed when the evidence demanded it

The first implementation used a linear-space heteroscedastic GP. Its likelihood honored the supplied uncertainties and its sampled functions could be differentiated, yet the latent state occupied all of $\mathbb{R}$. The complete data sweep located the practical consequence: 590 of 1,486 fits had a negative posterior median somewhere in the measured interval. Temperature and density are positive fields, so this was a support mismatch rather than an occasional tail event.

The positive map in Equation (6) repaired that specific defect without changing the additive measurement semantics. The same sweep then completed every profile with no negative posterior median or lower band. Figure 3 records the numerical diagnostic that forced the model decision.

The repair also changed what the app is willing to report. Under the positive model, 675 profiles had more than 90% of their radial scale length withheld; the plug-in linear baseline produced 170. The larger count follows the posterior sign gate, hyperparameter mixture, and endpoint exclusion. It expresses weaker evidence for a reciprocal gradient even when the underlying profile fit succeeds.

Synthetic functions provide a separate check because their exact derivatives are known. Twenty independent noise realizations were fitted for positive linear, peaked, and flat truths at 25 measurement locations. The table reports the fraction of evaluation radii at which a nominal 95% pointwise band contained the truth, averaged across repetitions.

Truth	f	reportable f'	reportable G_R
Linear	91.3%	93.8%	94.3%
Peaked	94.1%	96.3%	96.2%
Flat	95.2%	100.0%	100.0%

The flat truth gives the most important falsification: its inferred gradient interval contained zero over the entire grid, and the scale length was withheld everywhere. All synthetic positive-profile lower bands remained positive. Twenty repetitions are adequate for a compact implementation test; they do not estimate 95% coverage to production precision.

8 The application as an executable argument

The repository keeps the scientific maps in small modules. Their composition is

$$\begin{aligned}\texttt{data.py} \longrightarrow{} & \texttt{gp.py/positive\_gp.py} \\ & \longrightarrow \texttt{plotting.py} \longrightarrow \texttt{app.py}.\end{aligned} \tag{25}$$

`data.py` preserves the HDF5 (R, time) axes and units. `gp.py` defines the shared configuration, validity gate, result object, and labeled linear baseline. `positive_gp.py` evaluates Equations (9)–(14), draws the mixture, and performs Equations (17)–(20). `plotting.py` converts one posterior summary into aligned profile, gradient, and derived panels. `app.py` binds those maps to the shot, time, quantity, prior, boundary, and posterior controls.

Unit tests check the schema, ordered credible bands, derivative units, flat profile withholding, positive support, validity classification, distinct edge coordinates, optional data filtering, and the presence of multiple Laplace/quadrature components. The synthetic and all-profile validation scripts exercise a different axis: whether the assembled program behaves sensibly on known functions and on every supplied acquisition.

9 What has been established

The project constructs a coherent Bayesian path from processed MPTS points to positive spatial profiles and correlated gradient uncertainty. The heteroscedastic likelihood preserves each supplied error bar; the log latent field enforces the physical support; the Matérn covariance defines radial regularity; analytic differentiation carries the same posterior into f' and $d\log f/dR$; and the sign gate refuses to replace an unresolved zero denominator with a visually convenient scale length. The GUI exposes these choices as scientific controls and preserves geometric major radius in every label.

The present evidence supports a hackathon reconstruction tool and an exploratory scientific workflow. Four extensions separate it from a validated diagnostic product:

1. compare the Laplace/quadrature mixture with HMC or NUTS on strong, flat, outlier-contaminated, and absent-signal cases;
2. ingest upstream channel validity, saturation, over-range, and under-range status rather than relying on the provisional profile gate;
3. perform leave-one-channel-out and radial-block posterior predictive checks before adding a robust mixture likelihood; and
4. propagate an equilibrium ensemble before converting R derivatives into outward flux-surface gradients or a/L_f.

These are distinct validity axes. Successful optimization establishes numerical completion; positive support establishes one physical constraint; synthetic coverage probes interval calibration; and equilibrium mapping is required for transport interpretation. No one axis certifies the others.

Oral reconstruction

1. Starting from one HDF5 slice, reconstruct the complete map to a displayed $L_{f,R}$ sample and identify where measurement uncertainty enters.
2. Explain why $f = f_{\mathrm{ref}} e^z$ enforces positivity without imposing a lognormal likelihood on the processed measurements.
3. Derive $f' = fz'$ and use it to explain why $d\log f/dR$ is the primary normalized-gradient output.
4. State the posterior event that makes a finite $L_{f,R}$ scientifically unreportable, then distinguish it from endpoint derivative limitation.
5. Identify which evidence supports numerical completion, positive support, synthetic calibration, and flux-coordinate interpretation.

References

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